JAVA PROGRAM REPORT

Multilevel Inheritance in Java

Programming Language:	Java
Topic:	Multilevel Inheritance
Program File:	MultilevelInheritance.java

Objective

To demonstrate multilevel inheritance in Java, where EngineeringStudent inherits from Student, and Student inherits from Person. The object of EngineeringStudent can access methods defined in all three classes.

Java Source Code

```
class Person {
    void introduce() {
        System.out.println("I am a person");
    }
}

class Student extends Person {
    void study() {
        System.out.println("Student studies");
    }
}

class EngineeringStudent extends Student {
    void coding() {
        System.out.println("Engineering student learns coding");
    }
}

public class MultilevelInheritance {
    public static void main(String[] args) {

        EngineeringStudent e = new EngineeringStudent();

        e.introduce();
        e.study();
        e.coding();
    }
}
```

Program Explanation

1. Person Class: The base class defines introduce(), which prints “I am a person”.

2. Student Class: Student extends Person and adds the study() method. It also inherits introduce().

3. EngineeringStudent Class: EngineeringStudent extends Student, so it inherits both introduce() and study(). It adds coding().

4. Multilevel Inheritance: The inheritance chain is Person → Student → EngineeringStudent. Each level inherits accessible methods from the level above it.

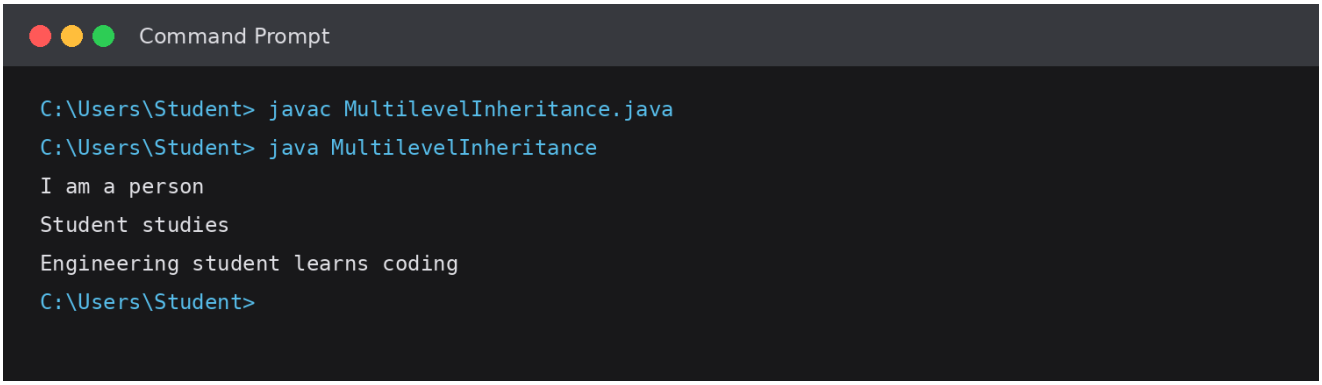
5. Main Method: An EngineeringStudent object is created, then introduce(), study(), and coding() are called in sequence.

Expected Output

```
I am a person
Student studies
Engineering student learns coding
```

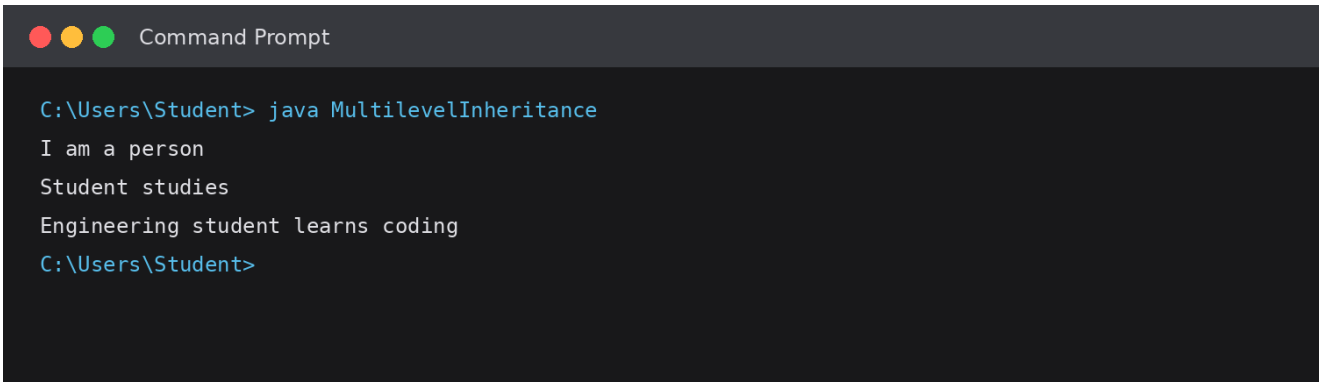
Execution and Output Screenshots

Test Case 1: Compilation and Execution



Verification: The program compiles and prints the inherited Person and Student messages, followed by the EngineeringStudent-specific message.

Test Case 2: Re-execution Verification



Verification: Re-running the class produces the same output sequence.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	All three methods execute in order.	Output matches expected messages.	**PASS**
Test Case 2	Same output on re-execution.	Output remains consistent.	**PASS**

Conclusion

The program successfully demonstrates multilevel inheritance. EngineeringStudent inherits methods through Student from Person and can use introduce(), study(), and coding(). The output confirms that methods from each level of the inheritance chain can be called on the EngineeringStudent object.